

Econ 30065: Microeconomics

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Class 29: Adverse Selection and the Job market

Interventions to improve efficiency

- Warranties and reputations – private solutions
 - Warranty – seller guarantees to take back the good if consumer is not satisfied (often for durable goods) **SIGNALLING**
 - Reputations – sellers try to build reputation over time that they have tested the good they sell and their quality is maintained over time/space
- Public interventions
 - Licenses (e.g. for doctors, health centers, pharmacies)
 - Mandatory checks to ensure quality and safety standards are maintained
- Remedies solve the moral hazard problem but still inefficiency

Example with warranty

Assume in the example of the cars with $MC_G=4$ and $MC_B=2$ it is possible to offer a warranty. Suppose the warranty costs 0.1 for owners of good car and 2.3 for owners of bad cars. Then the warranty solves the adverse selection problem.

$$P_w=4.4 \text{ (profits } 4.4-4-0.1=0.3>0)$$

$$P_{NW}=2.2 \text{ (profits } 2.2-2=0.2>4.4-2-2.3=0.1)$$

Example: signalling on the job market

- Things might change if type H could **signal** her ability
- But just saying "I am type H" is not a credible signal, as type L could easily say the same
- Signal must be *credible*
 - A signal is credible if sending it is convenient for type H, but not for type L
- In reality, higher education-costly action- serves (also) this purpose
- Indeed, besides higher productivity, higher ability also implies easier achievement of higher education levels (i.e less costly to get it for an H type!)
- How do conclusions change when signalling through education is possible?

Example: signalling

- Focus on case where worker proposes wage
- Before proposal is made, worker chooses whether to obtain a college degree
- Obtaining degree entails a disutility to worker (costly action)
- The disutility is decreasing in ability: C_H for type H and C_L for type L, with $C_L > C_H$
- Assume for simplicity that education does not affect productivity
 - It works only as a signal!

Example: signalling

- Ability still unobservable to firm
- But firm does observe worker's education level
 - ➔ Firm can condition decision to accept/reject worker's proposal on her education level
- We are interested in *separation*: this means type H goes to college but type L does not
- If separation happens, firm infers ability from education ➔ pay up to 100 for graduate
 - Type H proposes 100, firm accepts

Example: signalling

- When does separation occur?
- We need 2 constraint
 - Type H must find it preferable to graduate:
$$100 - C_H \geq 55$$
 - Type L must find it preferable NOT to graduate:
$$100 - C_L < 55$$
- ➔ Separation occurs if and only if:
$$C_L > 45 \geq C_H$$

Example: signalling

- When signaling opportunities are available and separation happens, highly productive worker behaves differently from less productive one
- Only type H chooses higher level of education
- While ability remains not observable directly, by observing worker's education choice in pre-contractual phase firm indirectly but perfectly infers worker's ability
- This is why firm accepts type H worker's proposal of 100!

signalling

- Signalling not as efficient as symmetric information because now the social cost is higher ($55+C$) → total surplus is lower.
- Note that this is only true because we assumed education and hard tasks not to have any real (productivity) value
 - If it were the other way, effect on total surplus would be *ambiguous*